

(Allow solidus notation and non SI units)

|     |      |       |                                                                                                                                                                                                     |     |
|-----|------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 2   | (a)  | (i)   | distance from a (fixed) point.....M1<br>in a specified direction ..... A1<br>(Allow 1 mark for 'distance in a given direction')                                                                     |     |
|     |      | (ii)  | (displacement from start is zero if) car at its starting position..... B1                                                                                                                           | [3] |
| (b) | (i)1 |       | $v^2 = u^2 + 2as$<br>$28^2 = 2 \times a \times 450$ (use of component of 450 scores no marks)..... C1<br>$a = 0.87 \text{ m s}^{-2}$ ..... A1<br>(-1 for 1 sig. fig. but once only in the question) | [2] |
|     |      | (i)2  | $v = u + at$ or any appropriate equation<br>$28 = 0.87t$ or appropriate substitution ..... C1<br>$t = 32 \text{ s}$ ..... A1                                                                        | [2] |
|     |      | (i)3  | $E_k = \frac{1}{2}mv^2$ ..... C1<br>$= \frac{1}{2} \times 800 \times 28^2$<br>$= 3.14 \times 10^5 \text{ J}$ ..... A1                                                                               | [2] |
|     | (i)4 |       | $E_p = mgh$ ..... C1<br>$= 800 \times 9.8 \times 450 \sin 5$ ..... C1<br>$= 3.07 \times 10^5 \text{ J}$ ..... A1                                                                                    | [3] |
|     |      | (ii)  | power = energy/time ..... C1<br>$= (6.21 \times 10^5)/32.2$ ..... C1<br>$= 1.93 \times 10^4 \text{ W}$ ..... A1<br>(power = $Fv$ with $F = mg \sin \theta$ scores no marks)                         | [3] |
|     |      | (iii) | some <u>work also done against friction</u> forces.....M1<br>location of frictional forces identified ..... A1<br>(allow reasonable alternatives)                                                   | [2] |

3 (a) (i) ductile B1